

Bounded-rank block projections and bounded-rank subsequences admit rank-one selections

Automated Math Discovery Run fa_banach_001
Model: GPT5.6

22 July 2026

Claimed status: substantial partial result, subject to human review. The results settle all uniformly bounded-rank block families and the infinite-subsequence question whenever one finite rank occurs infinitely often. The unbounded-rank and infinite-rank cases of Question 6.1 remain.

1 Source questions

Braga, Farah, and Vignati ask in Question 6.1 whether a sequence of projections $p_n \in C_u^*(X)$ whose every strong subseries belongs to $C_u^*(X)$ admits rank-one subprojections with the same property [1, Question 6.1, p. 20]. The exact source passage is reproduced below. The rank-two block case

6. ARE (I) AND (II) EQUIVALENT?

As mentioned earlier, the question whether the properties (I) and (II) of $C_u^*(X)$ discussed in the introduction are equivalent reduces to an innocent-looking question.

Question 6.1. Let X be a u.l.f. metric space. Suppose that p_n is a sequence of projections in $C_u^*(X)$ such that $\text{SOT-}\sum_{n \in M} p_n$ is in $C_u^*(X)$ for every $M \subseteq \mathbb{N}$ and each p_n is of rank strictly greater than 1.

1. Can we conclude that for every n there are projections $q_n \leq p_n$ of rank 1 such that $\text{SOT-}\sum_{n \in M} q_n$ is in $C_u^*(X)$ for every $M \subseteq \mathbb{N}$?
2. Can we at least conclude that there are an infinite $M \subseteq \mathbb{N}$ and $q_n \leq p_n$ of rank 1 for all $n \in M$ such that $\text{SOT-}\sum_{n \in M_0} q_n$ is in $C_u^*(X)$ for every $M_0 \subseteq M$?

A positive answer to either part of Question 6.1 would imply that (I) and (II) are equivalent. A partial result is given in [BFV21, Lemma 4.3], where a positive answer to (2) is given in case the projections of interest satisfy our usual regularity condition, that is, if $\inf_{n \in \mathbb{N}} \sup_{x \in X} \|p_n \chi_x\| > 0$.

Acknowledgments. This paper was written under the auspices of the American Institute of Mathematics (AIM) SQuaREs program and as spinoff of the ‘Ergodic theory and Banach algebras’ SQuaRE project, joint with

Figure 1: Question 6.1 in the source paper.

was restated as an explicit open question in Vignati’s 2026 habilitation [2, Question 5.31, p. 53]: if

$p = \text{SOT} - \sum_n p_n$ is a noncompact block ghost projection and every p_n has rank two, must there be rank-one $q_n \leq p_n$ such that $\text{SOT} - \sum_n q_n \in C_u^*(X)$?

Indeed, *bad* families of projections (i.e., those giving rise to ghosts) are exactly the cause for the failure of ONL, as the following show.

Theorem 5.29. *Let (X, d) be a u.l.f. metric space. The following are equivalent:*

- *There are no noncompact block ghost projections in $C_u^*(X)$.*
- *X has ONL for equi-approximable finite-rank projections.*

Theorem 5.30. *Let (X, d) be a u.l.f. metric space. The following are equivalent:*

- *There are no noncompact block-rank-one ghost projections in $C_u^*(X)$.*
- *X has ONL for equi-approximable rank-one projections.*

It is unknown whether the existence of a noncompact block ghost projection implies the existence of a noncompact block-rank-one ghost projection. A version of the following was asked a relatively long time ago, in [14].

Question 5.31. Let X be a u.l.f. metric space, and suppose that $p = \sum p_n$ is a noncompact block ghost projection in $C_u^*(X)$ where each p_n is of rank two. Are there rank one projections $q_n \leq p_n$ such that $q = \sum q_n$ is in $C_u^*(X)$?

Having a (positive) answer to Question [5.31] would significantly enhance our understanding of ghost projections, and consequently the K -theory of uniform Roe and quasi-local algebras.

Figure 2: The rank-two block question in the 2026 habilitation.

2 Definitions and scope

Let X be a uniformly locally finite metric space. A projection $p \in C_u^*(X)$ is a *block projection* if there are pairwise disjoint finite sets $X_n \subseteq X$ such that

$$p = \text{SOT} - \sum_{n \in \mathbb{N}} p_n, \quad p_n = \chi_{X_n} p \chi_{X_n}.$$

The p_n are mutually orthogonal finite-rank projections. It is *block-rank-one* when every nonzero p_n has rank one. An operator $a \in C_u^*(X)$ is a *ghost* if its matrix coefficients tend to zero at infinity. We use only two elementary facts: diagonal multipliers χ_A lie in $\ell_\infty(X) \subseteq C_u^*(X)$, and $C_u^*(X)$ is closed under continuous functional calculus.

3 Proof intuition

For rank two, some coordinate cut always distinguishes two orthogonal directions by a fixed amount. For a fixed higher rank, a compactness argument on isotropic measures on complex projective space supplies finitely many diagonal tests, one of which gives a uniformly isolated top eigenvalue in every block. Finite thresholding and continuous functional calculus then extract a rank-one subprojection from every block.

For a non-block Boolean family, an arbitrary finite-rank orthogonal sequence has a subsequence which is a summable perturbation of projections supported on pairwise disjoint finite coordinate sets. Boolean subseries survive that perturbation. The block theorem selects rank one, and a compactly perturbed partial isometry transports the selections back to the original projections.

4 The diagonal-gap lemma

Lemma 1 (uniform two-dimensional coordinate gap). *Let E be a two-dimensional subspace of $\ell_2(F)$, where F is finite, and let P be the orthogonal projection onto E . There is $A \subseteq F$ such that the two eigenvalues of $(P\chi_A P)|_E$ differ by at least $1/4$.*

Proof. Choose an orthonormal basis u, v of E . For $x \in F$, put

$$d_x = |u(x)|^2 - |v(x)|^2, \quad z_x = \overline{u(x)}v(x).$$

Then $\sum_x d_x = 0$ and $\sum_x z_x = 0$. For any $A \subseteq F$, the eigenvalue gap of $(P\chi_A P)|_E$ is

$$\left(\left| \sum_{x \in A} d_x \right|^2 + 4 \left| \sum_{x \in A} z_x \right|^2 \right)^{1/2}. \quad (1)$$

Set $D = \sum_x |d_x|$. If $D \geq 1/2$, take $A = \{x : d_x \geq 0\}$. Since $\sum_x d_x = 0$, $\sum_{x \in A} d_x = D/2 \geq 1/4$, and (1) gives the result. Suppose $D < 1/2$. Writing $a_x = |u(x)|^2$ and $b_x = |v(x)|^2$, we have

$$2|z_x| = 2\sqrt{a_x b_x} \geq a_x + b_x - |a_x - b_x|.$$

Thus $L := \sum_x |z_x| > (2 - 1/2)/2 = 3/4$. Averaging over $\theta \in [0, 2\pi]$ gives a θ such that

$$\sum_x \left| \text{Re}(e^{-i\theta} z_x) \right| \geq \frac{2L}{\pi}.$$

Let $A = \{x : \operatorname{Re}(e^{-i\theta} z_x) \geq 0\}$. Because $\sum_x z_x = 0$,

$$\left| \sum_{x \in A} z_x \right| \geq \frac{L}{\pi} > \frac{3}{4\pi}.$$

The second term in (1) makes the gap larger than $3/(2\pi) > 1/4$. □

5 Rank-two block selection

Theorem 2. *Let X be a uniformly locally finite metric space and let $p \in C_u^*(X)$ be a block projection with respect to pairwise disjoint finite sets $(X_n)_{n \in \mathbb{N}}$. Suppose every nonzero block $p_n = \chi_{X_n} p \chi_{X_n}$ has rank two. Then there are rank-one projections $q_n \leq p_n$ such that*

$$q = \operatorname{SOT}\text{-}\sum_{n \in \mathbb{N}} q_n \in C_u^*(X).$$

In fact, for every $M \subseteq \mathbb{N}$, $q_M = \operatorname{SOT}\text{-}\sum_{n \in M} q_n$ belongs to $C_u^(X)$.*

Proof. By Lemma 1, for every n there is $A_n \subseteq X_n$ such that the two eigenvalues

$$0 \leq \lambda_n^- \leq \lambda_n^+ \leq 1 \quad \text{of} \quad h_n = (p_n \chi_{A_n} p_n)|_{p_n \ell_2(X)}$$

satisfy $\lambda_n^+ - \lambda_n^- \geq 1/4$. Let $T = \{k/16 : 1 \leq k \leq 15\}$. For every n , choose $t(n) \in T$ so that

$$\lambda_n^- + \frac{1}{32} \leq t(n) \leq \lambda_n^+ - \frac{1}{32}. \quad (2)$$

Such a grid point exists because the interval between the two endpoints has length at least $3/16$. Put $I_t = \{n : t(n) = t\}$ and $A = \bigcup_n A_n$. For $I \subseteq \mathbb{N}$, set

$$p_I = \operatorname{SOT}\text{-}\sum_{n \in I} p_n = \chi_{\bigcup_{n \in I} X_n} p \chi_{\bigcup_{n \in I} X_n}.$$

Hence $p_I \in C_u^*(X)$. For every $t \in T$,

$$a_t = p_{I_t} \chi_A p_{I_t} = \operatorname{SOT}\text{-}\sum_{n \in I_t} p_n \chi_{A_n} p_n$$

also belongs to $C_u^*(X)$. Choose a continuous function $f_t : [0, 1] \rightarrow [0, 1]$ that is zero on $[0, t - 1/64]$ and one on $[t + 1/64, 1]$. The margins in (2) imply that

$$q^{(t)} := f_t(a_t)$$

is the block projection whose block in $p_n \ell_2(X)$, for $n \in I_t$, is the spectral projection of h_n associated with λ_n^+ . Thus every nonzero block of $q^{(t)}$ has rank one, and continuous functional calculus gives $q^{(t)} \in C_u^*(X)$. Since T is finite,

$$q = \sum_{t \in T} q^{(t)} \in C_u^*(X).$$

Writing $q_n = \chi_{X_n} q \chi_{X_n}$ yields rank-one $q_n \leq p_n$ and $q = \operatorname{SOT}\text{-}\sum_n q_n$. Finally, for every $M \subseteq \mathbb{N}$,

$$q_M = \chi_{\bigcup_{n \in M} X_n} q \chi_{\bigcup_{n \in M} X_n} \in C_u^*(X). \quad \square$$

6 The bounded-rank block extension

Lemma 3 (finitely many coordinate tests in fixed rank). *For every integer $r \geq 2$ there are $N_r \in \mathbb{N}$ and $\gamma_r > 0$ with the following property. Suppose $(F_n)_n$ are pairwise disjoint finite sets and $E_n \subseteq \ell_2(F_n)$ has dimension r . Writing P_n for the projection onto E_n , there are diagonal positive contractions $d_1, \dots, d_{N_r}$ on $\ell_2(\bigcup_n F_n)$ such that, for every n , some $(P_n d_j P_n)|_{E_n}$ has a simple largest eigenvalue whose distance from the second largest eigenvalue is at least γ_r .*

Proof. Identify an r -dimensional Hilbert space with $\mathbb{C}^r$. Let $\mathcal{K}_r$ be the weak-star compact space of positive measures μ on complex projective space $\mathbb{CP}^{r-1}$ satisfying

$$\int w w^* d\mu([w]) = I_r.$$

(The integrand is independent of the unit representative of $[w]$.) Fix $\mu \in \mathcal{K}_r$ and $[w_0] \in \text{supp}(\mu)$. Choose $0 < \varepsilon < 1/4$ and a continuous $f : \mathbb{CP}^{r-1} \rightarrow [0, 1]$, nonzero in $L_1(\mu)$, supported where $|\langle w, w_0 \rangle|^2 > 1 - \varepsilon$. Put

$$H_f(\mu) = \int f([w]) w w^* d\mu([w]), \quad m = \int f d\mu > 0.$$

The Rayleigh quotient at w_0 is at least $(1 - \varepsilon)m$. On $w_0^\perp$ it is at most εm , so the min-max principle gives

$$\lambda_1(H_f(\mu)) - \lambda_2(H_f(\mu)) \geq (1 - 2\varepsilon)m > 0.$$

The same strict inequality, with a smaller positive margin, holds on a weak-star neighborhood of μ . Compactness of $\mathcal{K}_r$ therefore gives finitely many functions $f_1, \dots, f_{N_r}$ and a common margin $\gamma_r > 0$.

For each n , fix a unitary $U_n : E_n \rightarrow \mathbb{C}^r$ and set $v_{n,x} = U_n P_n \delta_x$. The measure

$$\mu_n = \sum_{x: v_{n,x} \neq 0} \|v_{n,x}\|^2 \delta_{[v_{n,x}/\|v_{n,x}\|]}$$

belongs to $\mathcal{K}_r$. Because the F_n are disjoint, the prescriptions

$$d_j(x) = f_j([v_{n,x}/\|v_{n,x}\|]) \quad (x \in F_n, v_{n,x} \neq 0)$$

define global diagonal contractions. In the basis U_n , the compression $P_n d_j P_n$ is exactly $H_{f_j}(\mu_n)$, proving the claim. $\square$

Theorem 4 (uniformly bounded-rank block selection). *Let $p \in C_u^*(X)$ be a block projection with finite blocks p_n satisfying $2 \leq \text{rank}(p_n) \leq R$. Then there are rank-one $q_n \leq p_n$ such that $q_M = \text{SOT}-\sum_{n \in M} q_n$ belongs to $C_u^*(X)$ for every $M \subseteq \mathbb{N}$.*

Proof. Partition the blocks according to their rank $r \in \{2, \dots, R\}$. For each r , apply Lemma 3 and stitch its diagonal tests over the disjoint block supports. Thus each rank- r block has, for one of finitely many tests $d_{r,j}$, a top eigenvalue separated from the rest by γ_r .

Choose a finite grid $T_r \subset [0, 1]$ fine enough that whenever $\lambda_1 - \lambda_2 \geq \gamma_r$, some $t \in T_r$ satisfies

$$\lambda_2 + \gamma_r/4 \leq t \leq \lambda_1 - \gamma_r/4.$$

Partition the indices into the finitely many classes determined by (r, j, t) . If I is one class, then $p_I = \text{SOT}-\sum_{n \in I} p_n$ is a diagonal compression of p , hence is in $C_u^*(X)$, and $a_I = p_I d_{r,j} p_I \in C_u^*(X)$. A continuous function which is zero below $t - \gamma_r/8$ and one above $t + \gamma_r/8$ sends a_I to the projection onto its blockwise top eigenspaces. Those eigenspaces are one-dimensional. Summing over the finitely many classes gives $q = \text{SOT}-\sum_n q_n \in C_u^*(X)$. Finally, every q_M is a diagonal compression of the block projection q . $\square$

Corollary 5. *Every noncompact block ghost projection in $C_u^*(X)$ whose nonzero blocks have uniformly bounded finite rank contains a noncompact block-rank-one ghost projection. Consequently, Question 5.31 of [2] has an affirmative answer.*

Proof. Apply Theorem 4. The resulting q has infinite rank and is therefore noncompact. Since $0 \leq q \leq p$, for every $x, y \in X$,

$$|\langle q\delta_y, \delta_x \rangle|^2 \leq \langle q\delta_x, \delta_x \rangle \langle q\delta_y, \delta_y \rangle \leq \langle p\delta_x, \delta_x \rangle \langle p\delta_y, \delta_y \rangle.$$

The diagonal of the ghost projection p vanishes at infinity, so the display shows that q is a ghost. $\square$

7 Arbitrary Boolean families: an infinite subsequence

Lemma 6 (disjoint-support gliding hump). *Let $(p_n)_n$ be mutually orthogonal projections of the same finite rank r on $\ell_2(X)$. There are a subsequence $(p_{n_k})_k$, pairwise disjoint finite sets $F_k \subseteq X$, and rank- r projections $e_k \in \mathcal{B}(\ell_2(F_k))$ such that*

$$\sum_k \|p_{n_k} - e_k\| < \infty.$$

Proof. The orthogonality implies $p_n \rightarrow 0$ strongly. Hence, for every finite $F \subseteq X$, $\|p_n \chi_F\| \rightarrow 0$: strong convergence is uniform on the finite-dimensional unit ball of $\ell_2(F)$.

Inductively, after $F_1, \dots, F_{k-1}$ have been chosen, choose an unused n_k for which p_{n_k} is as small as desired on their union. Since p_{n_k} has finite rank, it is also as small as desired off some finite set G_k containing that union. Therefore the compression of p_{n_k} to $F_k = G_k \setminus \bigcup_{i < k} F_i$ is arbitrarily close to p_{n_k} . The spectral projection of this compression for $[1/2, 1]$ has rank r and can be made within 2^{-k} of p_{n_k} . Calling it e_k proves the summability assertion. $\square$

Theorem 7 (bounded-rank subsequence selection). *Let $(p_n)_n$ satisfy the hypotheses of Question 6.1. Suppose some finite rank $r \geq 2$ occurs for infinitely many p_n . Then there are an infinite $M \subseteq \mathbb{N}$ and rank-one $q_n \leq p_n$ for $n \in M$ such that $\text{SOT}-\sum_{n \in M_0} q_n \in C_u^*(X)$ for every $M_0 \subseteq M$. In particular, Question 6.1(2) has an affirmative answer whenever the ranks are uniformly bounded, and for every arbitrary-support rank-two family.*

Proof. Restrict to the infinite constant-rank subfamily and apply Lemma 6; relabel it as $(p_k)_k$, with block projections $(e_k)_k$. For every $M \subseteq \mathbb{N}$, the series

$$e_M - p_M = \sum_{k \in M} (e_k - p_k)$$

converges absolutely in norm. It is compact, so $e_M \in C_u^*(X)$. In particular $e = \text{SOT}-\sum_k e_k$ is a rank- r block projection in $C_u^*(X)$. Theorem 4 gives rank-one $s_k \leq e_k$ with every $s_M \in C_u^*(X)$.

For each k , let v_k be the canonical partial isometry from $e_k \ell_2(X)$ onto $p_k \ell_2(X)$ obtained from the polar decomposition of $p_k e_k$. By making the errors in Lemma 6 still smaller, the standard close-projections estimate gives $\sum_k \|v_k - e_k\| < \infty$. The initial projections of the v_k are orthogonal, as are their final projections, so $v = \text{SOT}-\sum_k v_k$ is a partial isometry. Moreover $v - e$ is an absolutely norm-convergent series of finite-rank operators; hence $v \in C_u^*(X)$. Put $q_k = v_k s_k v_k^*$. Then $q_k \leq p_k$ has rank one, and for every M ,

$$\text{SOT}-\sum_{k \in M} q_k = v s_M v^* \in C_u^*(X).$$

$\square$

8 Verification, limitations, and novelty

The script `code/verify_rank_two_gap.py` numerically enumerates all coordinate subsets for random finite rank-two subspaces and checks the $1/4$ gap bound. This is only a consistency check; Lemma 1 is the proof. The results do not settle Question 6.1 when the ranks tend to infinity or are infinite, and they do not give the all-indices conclusion for arbitrary non-block families. They do remove the complete bounded-rank block obstruction and settle the infinite-subsequence part whenever a finite rank occurs infinitely often. A bounded search on 22 July 2026 covered arXiv, the published EMS version, exact phrases from Question 6.1, combinations of uniform Roe, rank two, and rank-one subprojection, and Vignati's January 2026 habilitation. No prior proof of these selection statements was found; the habilitation still labels the rank-two block case Question 5.31. This is not an exhaustive novelty claim. Human review should focus on the projective-measure compactness lemma, the finite-threshold functional calculus, and the summable-perturbation transfer.

References

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- [2] A. Vignati, *Rigidity of Roe-like algebras*, Mémoire d'habilitation à diriger des recherches, 2026, Question 5.31, p. 53.